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# From Canonical Traces to Constrained Gain: A Practical Framework for Validating the HorrorPlace BCI System

This report provides a comprehensive analysis of the strategy for formalizing and validating the Rust Brain-Computer Interface (BCI) programming stack and the associated AI-chat code quality framework within the HorrorPlace ecosystem. The research goal is executed through a three-pronged approach that emphasizes empirical validation, balanced multi-objective optimization, and targeted mathematical formalization. This strategy prioritizes building a robust, deterministic system grounded in observable behavior before layering more abstract formal methods as a reinforcing safety net. The findings are structured around the core pillars of this strategy, detailing the specific research objects, proof targets, mathematical formulas, and governance mechanisms that define the path toward a provably safe and engaging BCI-driven horror experience.

## Pillar I: Empirical Validation as the Foundation for System Integrity

The foundational principle underpinning the entire research effort is the strategic prioritization of empirical validation over formal verification . This approach posits that a system's behavior must first be rigorously locked down and demonstrated to be correct under a wide array of realistic and synthetic conditions before more complex theoretical proofs are considered. The goal is to build a strong, verifiable baseline of deterministic behavior, using lightweight formal methods selectively to reinforce critical, high-risk components. This methodology is realized through a concerted effort to expand canonical data traces, strengthen mock drivers, and implement sophisticated test harnesses that serve as executable witnesses to the system's correctness.

A central component of this pillar is the expansion and utilization of canonical CSV/NDJSON traces . These files serve as the bedrock for replayability, allowing developers to deterministically reproduce system states and behaviors. The plan involves strengthening

the `BciMockDriver` to consume these traces and drive the system through various scenarios . This includes expanding the fixture library to cover not only typical gameplay but also boundary cases such as extreme input values, periods of steady overload, and rapid transitions between different states . By exhaustively testing against these canonical logs, the team can empirically verify that the system behaves as expected across its operational envelope. This empirical process is not merely about pass/fail testing; it extends to monitoring nuanced, long-run behaviors. For instance, the test harnesses are intended to assert cumulative metrics, such as the total time steps where a metric band like UEC or EMD exceeds an "overwhelmed" threshold (e.g., > 0.9), or the longest continuous run above that threshold . These assertions translate abstract safety requirements into concrete, measurable quantities that can be checked during Continuous Integration (CI) runs, providing a powerful mechanism for regression prevention .

The practical implementation of this empirical-first strategy is embodied in several key artifacts. The `Player Envelope + BCI Proof Loop PR bundle` exemplifies this approach by including a CI gate script, `ci/validate_envelope_policy.lua`, which acts as a worst-case excursion prover . This Lua script is designed to be integrated into CI pipelines (e.g., CircleCI, GitHub Actions) and takes as input policy JSON files. It then executes a series of predefined checks to validate whether the policy adheres to all necessary safety inequalities for stamina, sanity, battery, R accumulator, O accumulator, and noise budgets . The script returns a boolean result, failing the build if any constraint is violated, thus enforcing safety at the earliest possible stage of development . Similarly, the C-ABI harness, `bcimockharness.c`, serves as a critical executable witness . This tool drives the Rust-based `bcimockinit` and `bcimocknext` functions with canonical traces, asserting that the resulting RTPC outputs and metrics envelopes never violate their defined ranges ([0,1]) or DET-scaled limits . This cross-language harness provides a crucial link between the Rust numerics and the Unreal/C engine glue, offering empirical proof that the entire pipeline—from feature ingestion to final output—is behaving correctly for known-good inputs .

This empirical foundation is equally critical for governing AI-generated code. The research aims to move beyond ad-hoc reviews of AI output and instead define clear, quantitative metrics for success . The near-term target is to use replayed BCI logs and HorrorDirector mappings to establish integration tests . The performance of any AI-generated mapping or engine glue will be evaluated based on how well it respects the established guardrails—such as maximum tension per minute, maximum time above thresholds, and cooldown durations—when subjected to these canonical traces . This transforms the evaluation of AI contributions from a qualitative assessment into a quantitative one, directly measuring adherence to the core design principles of the BCI

system. The overarching implication of this pillar is that verifiability and determinism are paramount. By making empirical validation the default and primary means of assurance, the project builds a resilient and transparent system where behavior can be observed, tested, and trusted before more abstract, but potentially brittle, formal proofs are applied to reinforce specific high-risk areas.

| Component | Role in Empirical Validation | Key Artifact(s) |
|---|---|---|
| **Data Layer** | Provides deterministic inputs for replay and testing. Expands coverage to include boundary cases. | Canonical CSV/NDJSON traces , expanded fixtures |
| **Numerics & Logic** | Houses the core logic (EMA, mapping) that is driven by the mock driver. | `bcimappingengine` crate , `Ema` struct |
| **Engine Interface** | Acts as the bridge between the Rust logic and the game engine (Unreal/C). | C ABI (`bciloadcalibration`, `bcitickinputsample`, etc.) , `UAudioRtpcMappingComponent` |
| **Validation & Enforcement** | Executes tests, asserts properties, and enforces policies at runtime and in CI. | `bcimockharness.c` , `ci/validate_envelope_policy.lua` |

This layered approach ensures that while the system's day-to-day operation is guided by empirical evidence, the most critical aspects are fortified with formal guarantees where needed. This pragmatic balance allows for rapid iteration and confidence-building while systematically addressing the highest-priority safety concerns.

## Pillar II: The Balanced Multi-Objective Mutation-Gain Function for Safe Optimization

The second pillar of the research framework introduces a sophisticated optimization paradigm centered on the mutation-gain function, `G(f,p)`. This function is designed to navigate the inherent trade-offs between creating an engaging, fear-inducing player experience and maintaining strict adherence to safety and comfort constraints. Its architecture is a direct reflection of the project's dual mandate: to innovate in BCI-driven horror while ensuring player well-being remains non-negotiable. The core insight guiding its design is the explicit separation of hard feasibility constraints from soft optimization objectives, creating a robust framework that prevents undesirable outcomes by definition.

The most critical aspect of `G(f,p)` is its treatment of safety as a set of absolute, non-negotiable constraints . Any configuration or mutation that violates these hard boundaries receives zero or negative gain, irrespective of any potential improvements in engagement metrics . These hard constraints are explicitly defined as the DET caps,

comfort envelopes, and the subset rules dictated by the `contractCard`'s allowed metric bands . This establishes a strict "red line" that cannot be crossed in the pursuit of higher engagement. The function `G(f,p)` is therefore not just an optimizer but also a validator, ensuring that all proposed changes remain within a pre-defined safe operating region. This design choice directly addresses a common failure mode in AI-driven systems, where optimizing for a single objective (like engagement) can lead to catastrophic failures in other, unmodeled dimensions (like player distress) [9] .

For configurations that successfully satisfy all safety constraints, the gain function `G` is calculated as a weighted combination of engagement-related metrics . The primary metrics identified for this purpose are User Engagement (UEC), Emotional Impact/Dread (EMD), Ambiguity/Risk Resolution (ARR), and Cognitive Dissonance/Loss (CDL) . The formula effectively treats UEC and EMD as positive contributors to engagement, while ARR is included to encourage ambiguous and suspenseful experiences rather than purely action-oriented ones . CDL is treated as a risk term, capturing moments of cognitive dissonance that could lead to player discomfort or frustration, and is therefore penalized . The weights assigned to these metrics are not static; they are calibrated per gameplay tier and policy, allowing the system to adapt its optimization goals to different design contexts, such as a tense industrial corridor versus a surreal liminal space .

The conceptual framework for `G(f,p)` is being formalized into a precise mathematical expression. The goal is to create a normalized scalar value, likely in the range of [-1, 1] or [0, 1], that can be reliably compared across different mutations . A concrete formulation is proposed, which mathematically encodes the hard-constraint principle:

$$G(f,p)=\mathbf{1}\text{safe}\cdot(w_1\Delta\text{UEC}+w_2\Delta\text{ARR}-w_3\Delta\text{CDL})-w_4V$$

In this equation, `\mathbf{1}\{\text{safe}\}` is an indicator function that evaluates to 1 if all safety constraints are met and 0 otherwise, effectively nullifying the entire gain term if a violation occurs . The terms `ΔUEC`, `ΔARR`, and `ΔCDL` represent normalized changes in the respective entertainment metrics, and `V` is a penalty term representing violations of comfort or DET limits . The weights `w1`, `w2`, `w3`, and `w4` are tunable parameters that reflect the relative importance of each factor. This formalization makes the optimization problem explicit and analyzable.

This formalized `G` function serves a crucial role in the experimentation lifecycle. It is primarily used as an offline ranking metric to evaluate and compare different mutations, such as changes to mapping profiles, contract card definitions, or calibration parameters . Promotions of new, potentially superior configurations to production environments would be governed by Dead-Ledger policies that require two conditions to be met: first,

that the new configuration demonstrably satisfies all contractual safety bounds, and second, that it achieves a `G` score above a predefined threshold . This creates a rigorous, auditable process for innovation, where every change is held accountable not only for its potential to enhance the player experience but also for its unwavering commitment to safety. This framework provides a principled way to manage the complex interplay between fear, engagement, and player well-being, transforming subjective notions of "good" horror into a quantifiable, optimizable problem with clearly defined boundaries.

## Pillar III: Mathematical Formalization of Hex-Coded Mapping Families for Real-Time Stability

The third pillar of the research focuses on the mathematical formalization of the hex-coded mapping families (e.g., 0xPKLIN, 0xPKSIG, 0xPKHYS, 0xPKOSC), which are responsible for translating processed BCI signals and invariants into tangible game mechanics, primarily RTPC (Real-Time Parameter Control) values for audio and visual effects . The guiding principle for this formalization is a deliberate prioritization of local, real-time stability over global, long-horizon proofs of session-level player experience. This pragmatic approach ensures that the fundamental building blocks of the system are reliable and smooth on a per-tick basis, which is essential for immersion and player comfort, while deferring the more complex analysis of overall session trajectory to empirical evaluation.

The primary mathematical focus is on guaranteeing "real-time, frame-to-frame RTPC stability with bounded per-tick changes" at high frequencies, such as 60-120 Hz . Sudden, jarring jumps in audio volume, whisper intensity, or visual effects like flicker can break immersion and cause player discomfort. To prevent this, the formalization relies on tools that enforce smoothness and bounded behavior. The most important of these is the derivation of **Lipschitz bounds** for each mapping family . A function `f` is Lipschitz if there exists a constant `L` such that the distance between `f(x1)` and `f(x2)` is no more than `L` times the distance between `x1` and `x2`. Proving that a mapping family has a finite Lipschitz constant ensures that small changes in the input (BCI features, invariants) cannot produce arbitrarily large changes in the output (RTPCs), thereby preventing sudden jumps . This property is directly tied to the desired frame-to-frame stability.

In addition to theoretical bounds, practical enforcement is achieved through explicit constraints on per-tick changes. The system's design requires defining maximum allowable changes in RTPC values from one frame to the next . This acts as a hard

velocity limit, complementing the smoother behavior guaranteed by Lipschitz continuity. For stateful mapping families like hysteresis or oscillatory curves, the analysis becomes more complex. For hysteresis, the formalization must prove that, given the defined rules for switching states, the output cannot oscillate faster than a certain period, directly addressing constraints against strobing or high-frequency flicker . For oscillatory families, the formalization involves bounding the amplitude `A` and frequency `f` of a sinusoidal function (e.g., $y(t)=y_0+A\sin(2\pi ft)$) to ensure it remains within clamped bounds ($0\leq y(t)\leq 1$) under all circumstances . These mathematical derivations provide the theoretical justification for the hard-coded limits and clamps present in the implementation.

Crucially, this research plan wisely defers comprehensive long-horizon trajectory proofs to empirical evaluation via resurrection telemetry . Instead of attempting to create a single, perfect analytical model to predict the entire player's emotional journey, the plan is to treat the effectiveness of a mapping family as an empirical question. Mappings are evaluated based on their ability to maintain UEC and ARR within design bands while keeping CDL spikes and comfort violations low, as measured over many sessions . This empirical approach, guided by the mutation-gain function `G(f,p)`, is more adaptable and less brittle than relying solely on theoretical models. It acknowledges the complexity of the coupled human-computer system, where player fatigue, adaptation, and environmental factors make long-term prediction extremely difficult. By focusing on what happens *now* (frame-to-frame stability), the team can build a reliable foundation. The higher-level question of "what is the best long-term player experience?" becomes an empirical science, explored through controlled experiments and measured by the mutation-gain function, rather than a purely theoretical problem.

To ensure these mathematical properties are maintained in practice, the formalization is tightly coupled with schema enforcement and CI checks. Schemas for mapping families will include metadata fields specifying properties like maximum slope or allowed gain, which are derived from the mathematical analysis . A dedicated validator, running as part of the CI pipeline, will check that any new mapping implementation adheres to these bounds, preventing regressions before they are merged. This creates a closed-loop system where mathematical theory informs schema design, which in turn constrains implementation, and CI provides automated verification. This focused approach makes the formalization effort highly manageable and immediately useful, providing a solid guarantee of local stability upon which more complex, emergent behaviors can be built and safely measured.

# Core Artifacts: The Rust BCI Programming Stack and Its Proof Targets

The Rust BCI programming stack is composed of a series of interconnected crates and modules, each with distinct responsibilities and corresponding proof targets. The system is architected in layers, starting from offline data conversion, moving to a runtime smoothing and mapping engine, and finally interfacing with the game engine via a C Application Binary Interface (ABI). The overarching goal is to transform raw neurophysiological data into a stream of normalized, contractually bound entertainment metrics and RTPC controls, with proofs of correctness layered throughout the pipeline. The primary research objects are the `crates/bciconverter`, `crates/bci-ema-smoothing`, and the `bcimappingengine` crate within the main codebase .

At the lowest level is the **offline conversion layer**, represented by the `crates/bciconverter` crate, which exposes a command-line interface (CLI) named `hpc-bci-convert` . Its sole responsibility is to read various external EEG file formats (EDF, BDF, XDF, CSV) and convert them into a standardized interchange format: `.bci-features.ndjson` . Each line in this NDJSON file is a `bci-feature-envelope-v1` object. The proof targets for this component are primarily related to **format and privacy compliance**. Every emitted feature envelope must strictly validate against the `bci-feature-envelope-v1.json` schema, which uses `additionalProperties: false` to prevent the introduction of unexpected fields . Critically, the schema must deliberately omit any fields for raw sample arrays or personally identifiable information (PII) such as subject IDs or demographic data . The "proof" here is a combination of structural validation (schema compliance) and procedural enforcement (anonymization policies in the converter contract), supported by CI jobs that lint for forbidden keys .

The **runtime numerics layer** is the heart of the stack and consists of two main components: the `crates/bci-ema-smoothing` library (compiled as `libhpcbciema.so`) and the `bcimappingengine` crate in the main `Codebase-of-Death` repository . The `libhpcbciema` library is exposed via a single C ABI function, `hpcbciprocess`, which serves as the central processing unit for BCI data . This function takes a `bci-feature-envelope-v1` and a contract context JSON as input, applies EMA filtering and calibration, and outputs a fully schemavalid `bci-metrics-envelope-v1` . The proof targets for this layer are far more stringent and mathematical. They include: 1. **EMA Stability and Range Preservation:** The core EMA filter, defined by the recurrence relation $v_t = \alpha x_t + (1-\alpha) v_{t-1}$, must be proven to preserve bounds . Specifically, for any input sequence $x_t \in [0,1]$ and initial value $v_0 \in [0,1]$, the output $v_t$ must remain within $[0,1]$ for all time `t` . This is typically proven by induction and forms a

fundamental safety property . Furthermore, the step response and settling time should be analyzed as a function of the smoothing parameter $alpha$ to ensure appropriate responsiveness without excessive jitter or lag . 2. **Contract-Bounded Outputs:** The output metrics envelope's bands for UEC, EMD, STCI, CDL, and ARR must be proven to remain within the `[0,1]` range, and the computed `detEstimate` must stay within its `[0,10]` cap, for any valid input and legal contract context . 3. **DET-Aware Safety Bounds:** If the contracts specify safety caps where metric bands are gated by the DET level (e.g., $m_i \leq g_i(\text{DET})$), the implementation must formally enforce these inequalities . This often involves clamping the output metrics after computation to ensure they never exceed the DET-dependent ceiling .

The **mapping engine** within `bcimappingengine` is responsible for taking the smoothed metrics and invariants from the `InputSample` and transforming them into RTPC targets for the game engine . The next major objective is to replace stub linear mappings with a proper registry of hex-coded curve families (0xPKLIN, 0xPKSIG, etc.), each with its own pure mapping function and associated safety envelope . The proof targets for this component mirror those for the numerics layer but are focused on the final outputs: 1. **RTPC Range and Safety:** The generated RTPC values must always remain within their defined `[0,1]` range and respect any DET-scaled caps specified in the active contract . 2. **Frame-to-Frame Stability:** As detailed in Pillar III, the mapping functions must be analyzed for Lipschitz continuity and bounded per-tick changes to ensure smooth, stable modulation of audio and visual effects at 60-120 Hz . 3. **Stateful System Stability:** For hysteresis and oscillatory families, the proofs must demonstrate that outputs remain bounded and do not exhibit uncontrolled oscillations or stick in extreme states under all admissible inputs .

Finally, the **engine glue layer**, comprising Unreal C++ components and Lua scripts, is proven mainly by **conformance** . These components act as wrappers around the C ABI, passing invariants and metrics from engine APIs and calling the Rust functions. Their "proof" lies in demonstrating that they only call the approved C ABI, never bypass contract enforcement, and do not implement additional numerics themselves. This is enforced by code review and CI lints that scan for forbidden device SDK calls or unsupported APIs .

| Layer | Primary Research Object | Key Proof Target(s) |
| --- | --- | --- |
| **Offline Conversion** | `hpc-bci-convert` CLI (`bciconverter` crate) | Schema compliance (`bci-feature-envelope-v1.json`), PII/raw data removal, format validity . |
| **Runtime Numerics** | `hpcbciprocess` (`libhpcbciema`) | EMA range preservation ([0,1]), contract-bounded metric outputs ([0,1][5]), DET-cap enforcement . |
| **Mapping Engine** | `bcimappingengine` crate | Contract-bounded RTPC outputs ([0,1]), frame-to-frame stability (Lipschitz bounds), bounded oscillation/sticking . |
| **Engine Glue** | Unreal C++ / Lua adapters | Conformance to C ABI, no bypassing of contract enforcement, no implementation of numerics . |

This layered structure, with progressively stronger proof targets at each stage, ensures that errors are caught early and that the final connection to the game engine is both safe and predictable.

## Governance Layer: Ensuring AI-Generated Code Adheres to BCI Contracts

The governance of AI-generated code for the BCI system is managed through a rigid, schema-driven framework designed to channel the AI's capabilities within a predefined set of rules. This approach treats the AI not as an autonomous agent but as a powerful assistant operating within a tightly constrained environment, ensuring that all generated artifacts are safe, correct, and auditable. The research focuses on turning high-level playbook rules into measurable behaviors that can be enforced by tooling and CI, thereby creating a robust defense-in-depth strategy for maintaining code quality.

The foundation of this governance layer is the **schema-first doctrine**, codified in documents like the `AI-BCI-PLAYBOOK.md` and `BCI-Dev-Guide.md` . These documents define the immutable law for BCI-related work, mandating strict adherence to schemas like `bci-feature-envelope-v1.json` and `bci-metrics-envelope-v1.json` . The AI is instructed to generate code that only ever reads and writes these schematized envelopes, never creating ad-hoc JSON structures. The proof of this adherence comes from CI jobs that validate all generated NDJSON and JSON artifacts against their corresponding schemas, failing the build on any unknown or forbidden properties . This provides a structural proof that constellation logs and BCI data streams remain clean, derived, and anonymized, free of raw EEG or PII .

To make these constraints actionable for an AI, the framework relies on a suite of custom tooling designed to automate validation and enforce contracts. One key objective is the creation of a **Rust "schema helper" crate** that generates typed structs from the BCI schema IDs . This allows AI-generated Rust code to operate at the type level, significantly improving correctness and reducing the likelihood of errors compared to manipulating raw JSON objects. Similarly, Lua scripts like `hpcbciimport.lua` and `hpcbciadapter.lua` are provided with small, tested helpers for decoding envelopes, so the AI only needs to fill in the logical glue rather than reimplementing field checks . For broader governance, a `bci-contract-lint` tool is envisioned to validate the content of converter contracts, ensuring that anonymization policies are correctly set and licenses permit the use of derived features . These tools lower the barrier for the AI to comply with the rules by providing it with the right abstractions and automated validators.

A second critical governance mechanism is the enforcement of **narrow API discipline**. The AI is restricted to using a small set of approved engine-facing APIs, such as `EEGFeatureService` in Unity, `UEEGFeatureSubsystem` in Unreal, and the `hpcbciadapter.lua` module . The rationale is to prevent the AI from inventing new interfaces or, worse, touching vendor SDKs like BrainFlow or LSL directly from game-tier code . The proof of this discipline is made measurable by defining an **API adherence score** for AI-generated code . This score is calculated based on the ratio of calls to approved entry points versus forbidden symbols. A static checker is developed for each language (C#, C++, Lua) that flags any use of disallowed names in engine-tier code, and the results of these checks become a quantitative metric tracked by CI . This transforms a vague rule into a concrete KPI that can be monitored and improved over time.

The third pillar of governance is the use of **prism envelopes and manifests** to enforce contract discipline and traceability . Every AI-authored change must be packaged within a `ai-authoring-response-v1` envelope, which contains metadata such as `targetRepo`, `targetPath`, `schemaRef`, and `tier` . This wrapper, itself conforming to a `prism-envelope-v1` schema, ensures that every change is explicitly scoped, traceable, and auditable . The proof of compliance is verified by a `validatemanifests.rs` module that runs in CI, checking for things like one-file-per-request rules, the presence of required fields, and the inclusion of a Dead-Ledger reference when necessary . A **Prism Discipline score** can be defined to measure the percentage of responses that adhere to these manifest rules, providing another quantitative gauge of AI behavior .

Finally, the ultimate judge of AI-generated code is the **HorrorDirector and replay traces**. The framework moves beyond static validation to dynamic, empirical testing.

Replay regression tests are designed to feed canonical BCI logs into the HorrorDirector and assert that the resulting tension and pacing trajectories respect all guardrails, such as maximum rate of change, cumulative time above an overwhelm threshold, and longest continuous run length . An AI-generated mapping config is deemed invalid if it causes the system to violate these runtime constraints. This creates a powerful feedback loop: the AI generates a mapping, the system empirically tests it against a corpus of real-world data, and the test results determine its fitness. This approach grounds the entire process in observable reality, ensuring that even the most theoretically sound AI generation is ultimately validated by its performance in the simulated game environment.

## Synthesis: A Pragmatic Path from Schema to Safe, Engaging Horror Experiences

The research initiative detailed herein outlines a comprehensive and pragmatic strategy for formalizing and validating the Rust BCI programming stack and its surrounding AI-assisted development workflow. The synthesis of the three core pillars—empirical validation, balanced multi-objective optimization, and targeted mathematical formalization—reveals a mature and robust engineering philosophy. This approach does not seek to achieve an unattainable state of total formal verification from the outset. Instead, it builds a resilient system on a foundation of empirical certainty, layering formal methods and mathematical rigor only where they provide the most significant benefit, specifically for high-risk, critical-path components. This hybrid strategy provides a clear and actionable roadmap from abstract design principles to a concrete, verifiable, and safe implementation.

The first pillar, prioritizing empirical validation, establishes the system's bedrock of trustworthiness. By leveraging canonical traces, mock drivers, and CI-integrated harnesses, the team can deterministically explore the system's behavior under a vast array of conditions . This empirical-first approach ensures that every component's function is grounded in observable reality before theoretical proofs are considered. It transforms abstract safety requirements into concrete, measurable assertions that can be automatically checked, creating a powerful and repeatable safety net. This same empirical lens is applied to AI-generated code, where performance is judged not by its syntactic correctness alone, but by its behavioral fidelity to established guardrails when exercised against replayed BCI logs .

The second pillar, the balanced multi-objective mutation-gain function, `G(f,p)`, provides the intellectual framework for navigating the delicate trade-off between engagement and safety. Its design, which separates non-negotiable safety constraints from soft optimization objectives, is a direct solution to the classic "optimization trap" where maximizing one metric leads to failure in another [9]. By defining a formalized gain function that is nullified by any safety violation, the system embeds ethical and safety considerations directly into its optimization calculus . This transforms the design of BCI-driven horror from an art of subjective guesswork into a science of principled, constrained optimization, where every innovation is held accountable for both its potential to engage and its unwavering commitment to player well-being.

The third pillar, focusing mathematical formalization on real-time stability, demonstrates a keen understanding of the practical challenges of controlling complex systems. By concentrating on proving frame-to-frame RTPC stability and bounded per-tick changes for hex-coded mapping families, the research ensures that the user experience is smooth, immersive, and comfortable at a fundamental level . This local stability is a prerequisite for a good overall experience. Deferring the more complex analysis of long-horizon session trajectories to empirical evaluation via resurrection telemetry is a wise and pragmatic decision . It acknowledges the inherent unpredictability of human-computer interaction over extended periods and embraces an empirical, data-driven approach to understanding and shaping the emergent player experience. This mirrors modern control theory, where local stability is ensured first, and then global behavior is shaped through feedback loops and optimization.

Together, these pillars are supported by a rigid, schema-driven governance layer that treats AI not as a black box but as a tool operating within a tightly defined contract. Through the use of custom tooling, narrow APIs, and prism-enveloped authoring, the system ensures that all code, whether written by humans or AI, conforms to a shared set of principles of correctness, safety, and auditability . This comprehensive framework represents a significant step forward in the engineering of interactive, biometrically-controlled systems. It provides a blueprint for building not just more engaging games, but safer, more reliable, and more trustworthy ones.

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